

Integer solutions in feasible regions



1

- a $(6, 12), (7, 12), (8, 12), (8, 11), (8, 10), (7, 11)$
- b $(0, 2), (1, 2), (2, 2), (0, 1), (1, 1), (0, 0), (1, 0)$
- c $(5, 6), (5, 5), (5, 4)$
- d $(-6, -7), (-6, -6), (-6, -5), (-5, -5), (-5, -4), (-5, -3)$
- e $(6, 7), (5, 6), (6, 6), (7, 6), (5, 5), (6, 5), (7, 5), (6, 4)$

2

- a
 - i $(0, 3), (1, 3), (0, 2), (1, 2), (0, 1), (0, 0)$
 - ii $(0, 3)$
- b
 - i $(5, 7), (5, 6), (5, 5), (5, 4), (6, 5)$
 - ii $(5, 4)$
- c
 - i $(6, 7), (5, 6), (6, 6), (7, 6), (5, 5), (6, 5), (7, 5), (6, 4)$
 - ii $(5, 6)$

3

- a
 - i $(2, 4), (3, 4), (4, 4), (5, 4), (6, 4), (2, 3), (3, 3)$
 - ii $(2, 3)$
- b
 - i $(-6, -4), (-5, -4), (-6, -5), (-5, -5), (-4, -6)$
 - ii $(-5, -4)$



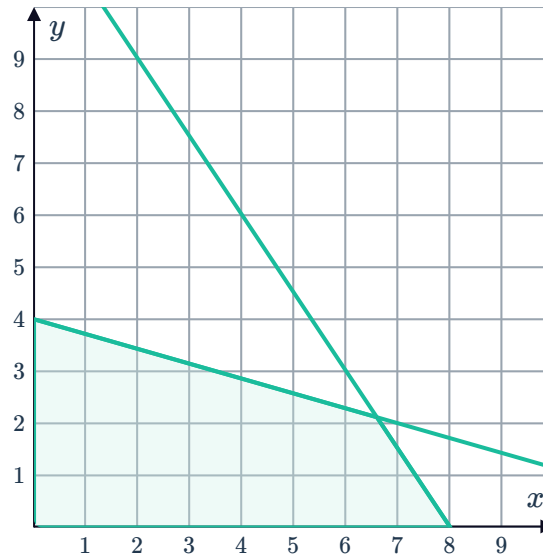
Integer solutions in linear programming applications

4

a A number of students must be a whole number as it is not possible to have a part of a student.

b $x \geq 0, y \geq 0, 3x + 2y \leq 24$ and $x + 3.5y \leq 14$

c



d $P = 200x + 220y$

e

Possible solutions	Profit (dollars)
(0, 4)	880
(7, 0)	1400
(6, 2)	1640

f \$1640

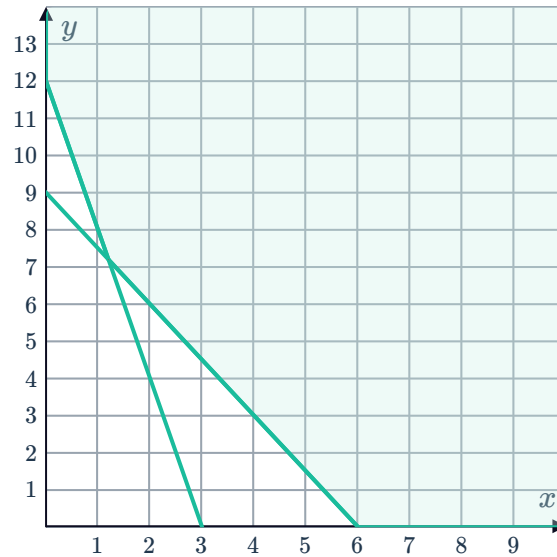
g 6 English students and 2 Mathematics students

5

a $x \geq 0, y \geq 0, 8x + 2y \geq 24$ and $6x + 4y \leq 50$



b



c $C = 25.8x + 15.2y$

d $(1, 8), (2, 6), (0, 12), (6, 0)$

e

Possible solutions	Ingredients cost (\$)
$(0, 12)$	182.40
$(6, 0)$	154.80
$(1, 8)$	147.40
$(2, 6)$	142.80

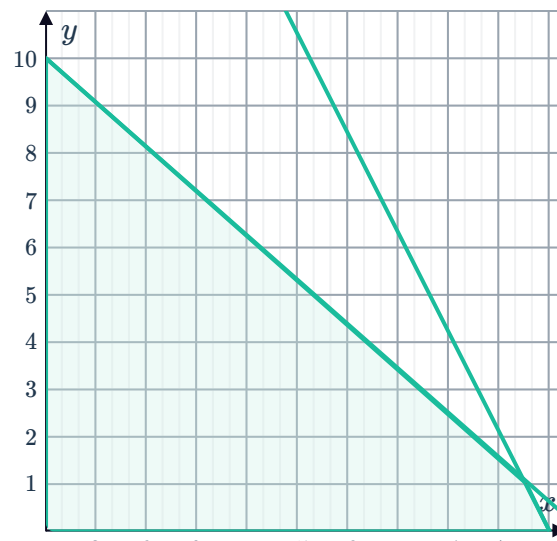
f \$142

g 2 cakes and 6 muffins

6

a $x \geq 0, y \geq 0, 175x + 250y \leq 5250$ and $0.25x + 0.8y \leq 8$

b



c $P = x + y$



d $(0, 10), (25, 2), (28, 1), (30, 0)$

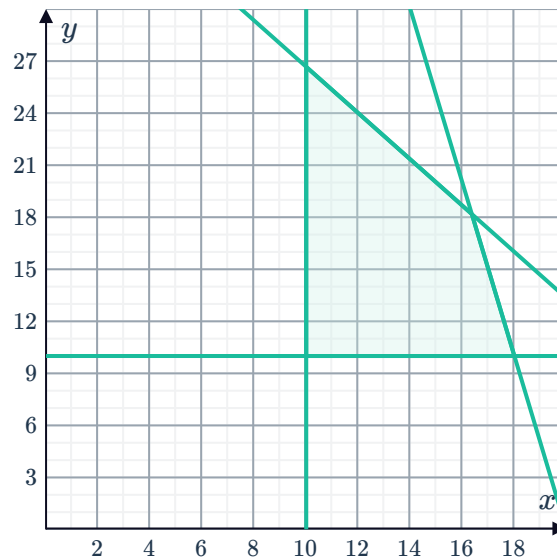
e 30 plants

f 30 carrots and 0 parsnips

7

a $x \geq 10, y \geq 10, 50x + 10y \leq 1000$ and $20x + 15y \leq 600$

b



c $V = 5x + 2y$

d $(16, 18), (10, 26), (18, 10)$

e

Possible solutions	Value (\$)
$(16, 18)$	119
$(0, 26)$	102
$(18, 0)$	110

f \$116

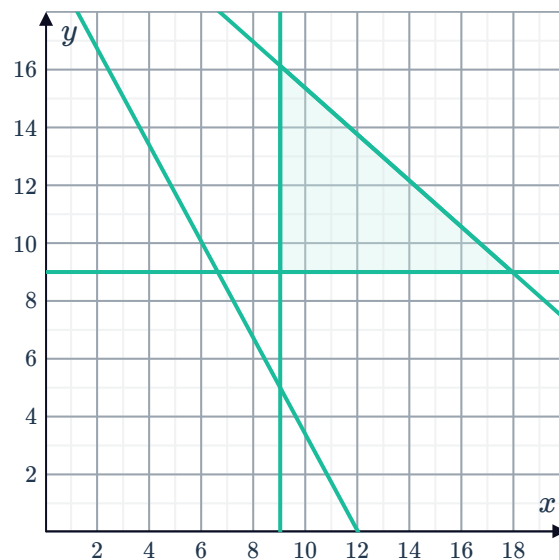
g 16 tomato plants and 18 strawberry plants

8



a $x \geq 9, y \geq 9, 25x + 15y \geq 300$ and $12x + 15y \leq 350$

b



c $V = 15x + 12y$

d $(18, 9), (9, 16), (9, 9)$

e

Possible solutions	Value (\$)
$(18, 9)$	378
$(9, 16)$	327
$(9, 9)$	243

f \$378

g 18 protein powder sachets and 9 packets of pancake mix